

Figures and Diagrams

Figure 1: Energy Scale Hierarchy Across Physical Regimes

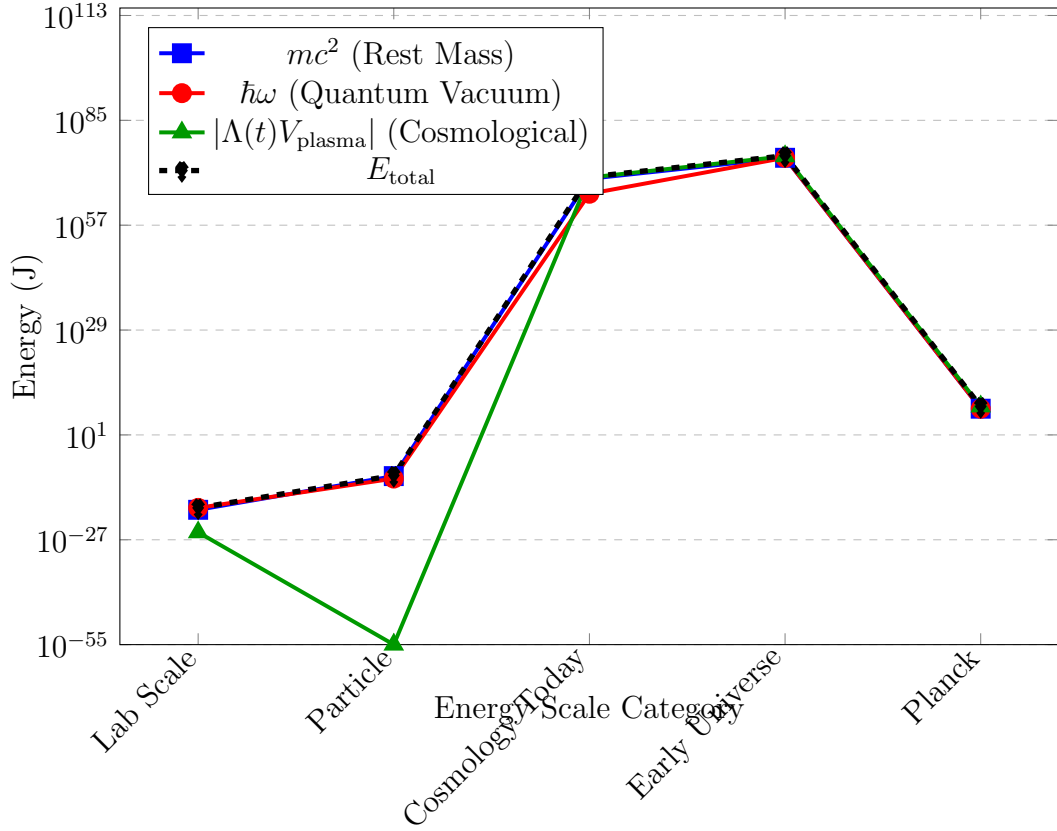


Figure 1: Energy contributions from the three terms in equation $E_{\text{total}} = mc^2 + \hbar\omega + \Lambda(t)V_{\text{plasma}}$ across different physical scales. At laboratory scales, quantum vacuum energy dominates. At particle physics scales, rest mass dominates. At cosmological scales today, matter and dark energy are comparable. In the early universe, all three terms are of similar magnitude, enabling near-perfect cancellation.

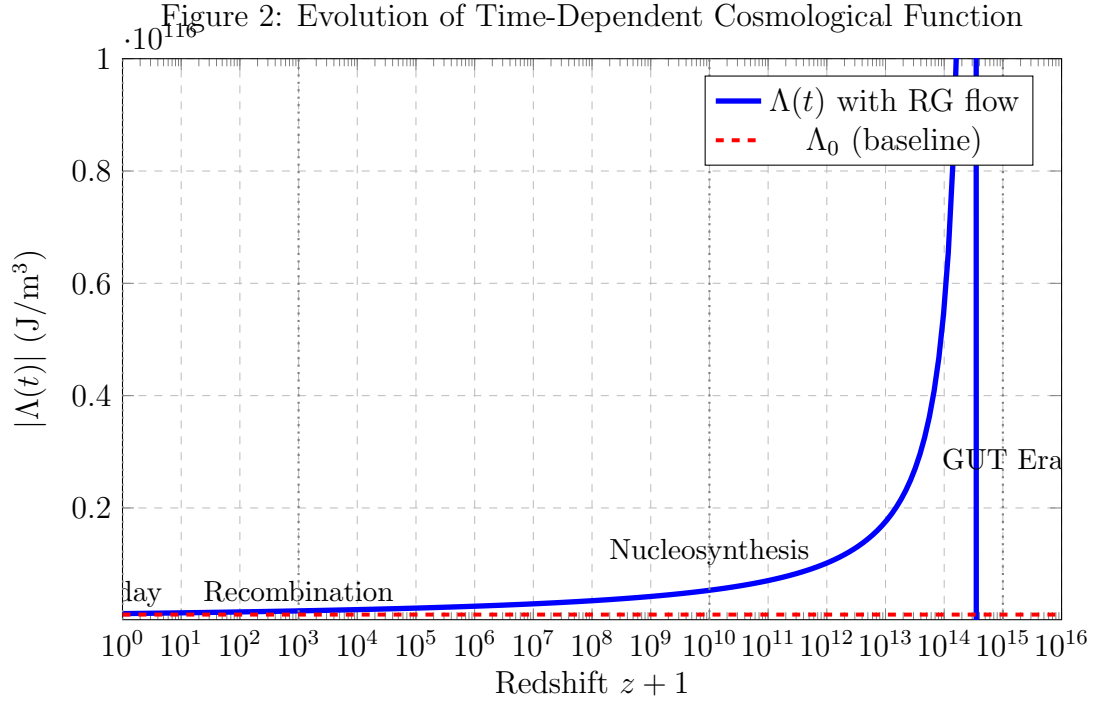


Figure 2: Time-dependence of the cosmological function $\Lambda(t)$ from the present epoch ($z = 0$) back to the GUT era ($z \sim 10^{15}$). The function evolves through renormalization group flow and QCD corrections, increasing in magnitude toward earlier times. The baseline Λ_0 (red dashed line) represents the zero-temperature Casimir contribution from M5-branes.

Figure 3: Energy Cancellation Mechanism Over Cosmic History

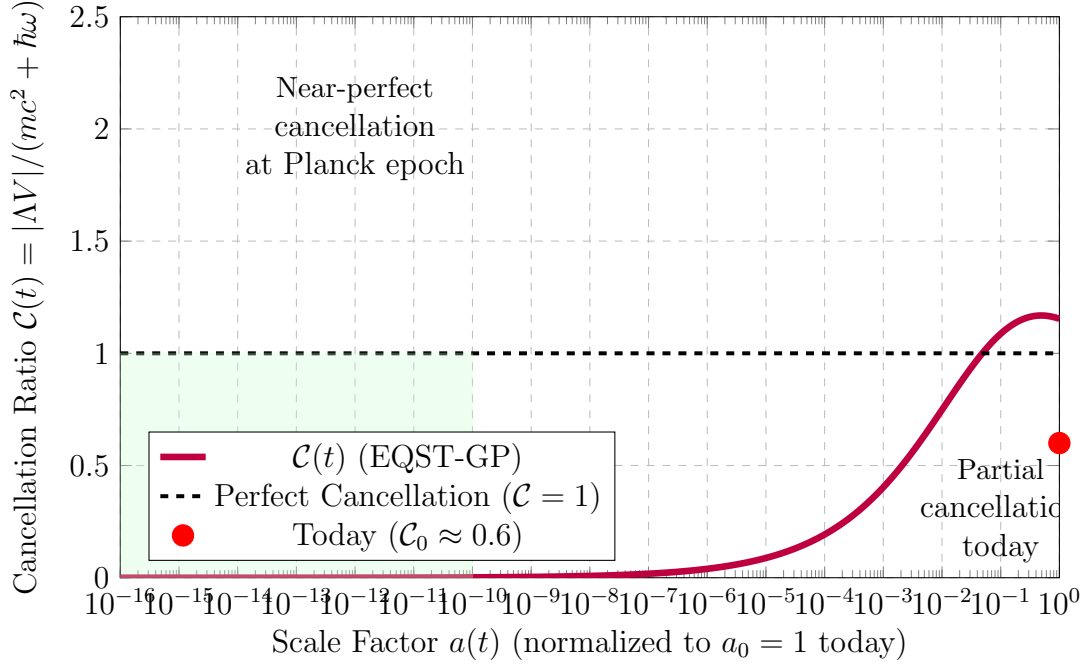


Figure 3: Evolution of the cancellation ratio $\mathcal{C}(t) = |\Lambda(t)V_{\text{plasma}}|/(mc^2 + \hbar\omega)$ from the Planck epoch to today. Near-perfect cancellation ($\mathcal{C} \approx 1$, shaded green region) occurs in the very early universe, explaining why extreme energy densities did not cause immediate collapse or runaway expansion. As the universe expands, cancellation becomes less complete, reaching $\mathcal{C} \approx 0.6$ today.

Figure 4: Cosmological Energy Density Evolution

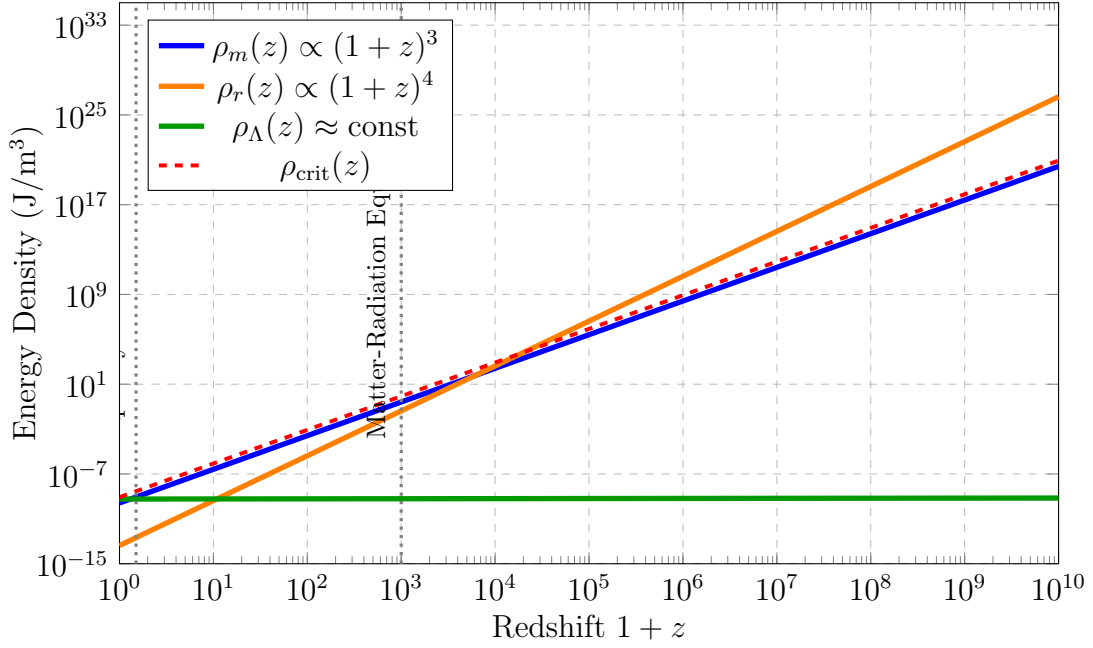


Figure 4: Evolution of energy density components with redshift in standard Λ CDM cosmology augmented with EQST-GP time-dependent dark energy. Matter density scales as $(1+z)^3$, radiation as $(1+z)^4$, while dark energy remains approximately constant (with slow logarithmic evolution). Matter-radiation equality occurs at $z \sim 3400$. Matter-dark energy equality (onset of acceleration) occurs at $z \sim 0.3$.

Figure 5: Kaluza-Klein Mass Tower for $R_{\text{KK}} = 10l_P$

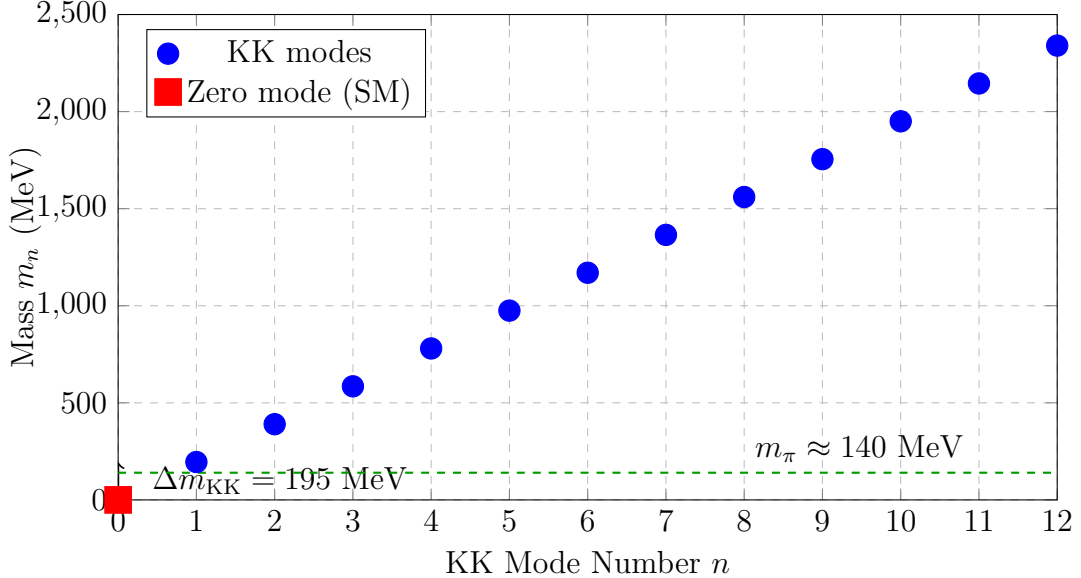


Figure 5: Kaluza-Klein mass spectrum for compactification radius $R_{\text{KK}} = 10l_P \approx 1.6 \times 10^{-34}$ m. The mass gap $\Delta m_{\text{KK}} = \hbar/(R_{\text{KK}}c) \approx 195$ MeV explains why extra dimensions are not directly observable at low energies. The first KK mode lies just above the pion mass, making detection challenging in typical nuclear/hadronic processes.

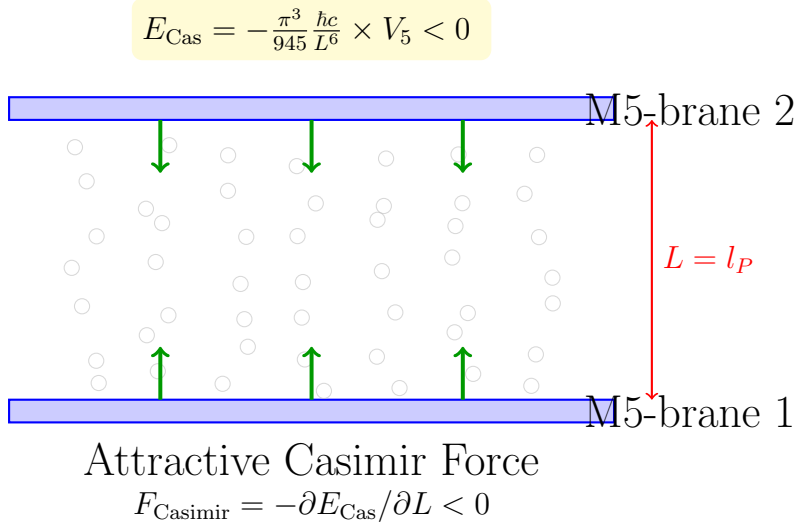


Figure 6: Schematic representation of Casimir effect between parallel M5-branes separated by distance $L = l_P$ in compactified dimensions. Vacuum fluctuations (gray circles) between the branes generate attractive force (green arrows) due to restricted mode spectrum. This negative Casimir energy provides the baseline Λ_0 in the EQST-GP framework.

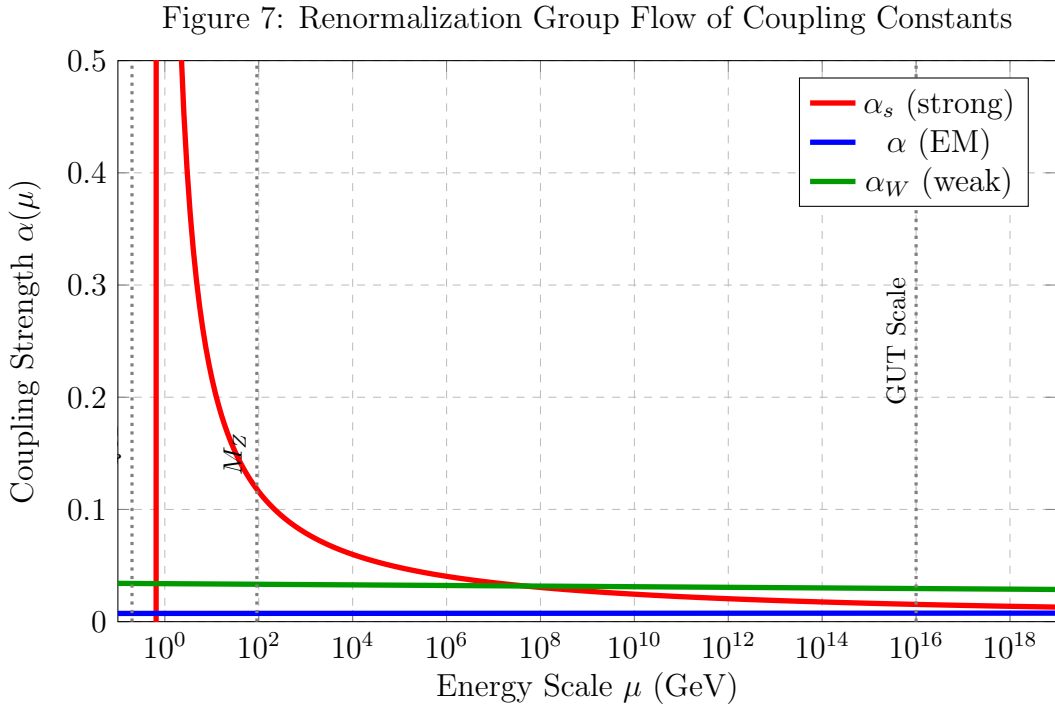


Figure 7: Running of gauge coupling constants with energy scale μ according to renormalization group equations. The strong coupling α_s exhibits asymptotic freedom, decreasing toward high energies and diverging near $\Lambda_{\text{QCD}} \sim 200$ MeV. Electromagnetic and weak couplings increase with energy. All three couplings approach similar values near the GUT scale $\sim 10^{16}$ GeV, suggesting grand unification.

Figure 8: Thermal History and Phase Transitions of the Universe

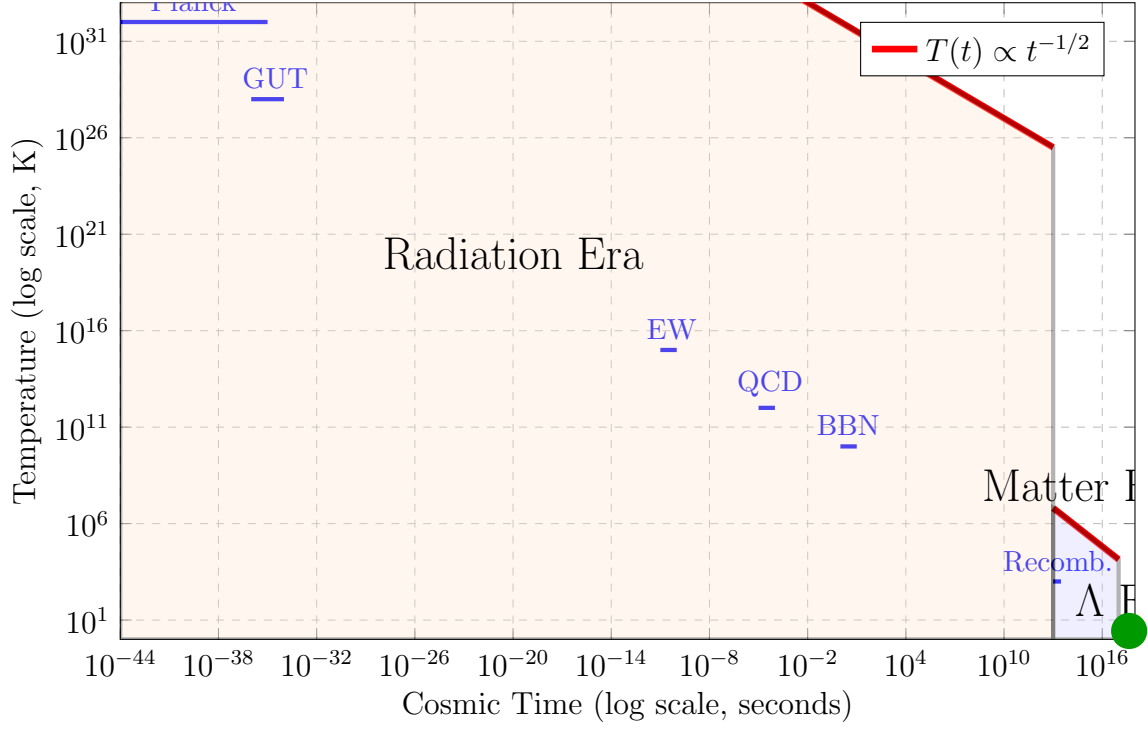


Figure 8: Thermal history of the universe showing temperature evolution and major phase transitions. Temperature decreases as $T \propto t^{-1/2}$ in radiation-dominated era, $T \propto t^{-2/3}$ in matter-dominated era, and approaches constant in dark energy era. Key transitions marked: Planck epoch ($T \sim 10^{32}$ K), GUT phase transition ($T \sim 10^{28}$ K), electroweak symmetry breaking ($T \sim 10^{15}$ K), QCD confinement ($T \sim 10^{12}$ K), Big Bang nucleosynthesis ($T \sim 10^{10}$ K), and recombination ($T \sim 10^3$ K). The EQST-GP framework predicts enhanced $|\Lambda(t)|$ at higher temperatures.